

MH1301 Discrete Mathematics

Tutorial 6

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Overview

This document covers the Week 7 tutorial-discussion set from the handout. The main themes are:

- solving coupled recurrence systems by elimination and matrix methods;
- higher-order linear homogeneous recurrences with distinct roots and repeated roots;
- non-homogeneous recurrences solved by undetermined coefficients, annihilators, and sequence transformations;
- derivation of classical summation formulas from recurrence relations;
- structural properties of recurrence-defined sequences, including induced recurrences and closure under linear combination.

Question map.

- Additional Exercise 1: simultaneous recurrences in two sequences.
- Ex. 8.2.11 and Ex. 8.2.15: third-order homogeneous recurrences.
- Ex. 8.2.22 and Ex. 8.2.25: non-homogeneous recurrences with polynomial and exponential forcing terms.
- Ex. 8.2.26: sum of squares via recurrence methods.
- Additional Exercise 2: a higher-order recurrence satisfied by Fibonacci numbers.
- Additional Exercise 3: linear combinations of solutions.

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Question 1 (Additional Exercise 1: Simultaneous recurrence relations)

Problem

Solve the simultaneous recurrence relations

$$a_n = 3a_{n-1} + 2b_{n-1}, \quad b_n = a_{n-1} + 2b_{n-1},$$

with

$$a_0 = 1, \quad b_0 = 2.$$

Solution

Method 1: Eliminate one sequence

We first eliminate a_n to obtain a recurrence purely in b_n .

From

$$b_n = a_{n-1} + 2b_{n-1},$$

we get

$$a_{n-1} = b_n - 2b_{n-1}. \quad (1)$$

Replacing n by $n + 1$ in (1) gives

$$a_n = b_{n+1} - 2b_n. \quad (2)$$

Now substitute (1) and (2) into

$$a_n = 3a_{n-1} + 2b_{n-1}.$$

Using (2) on the left-hand side and (1) on the right-hand side,

$$b_{n+1} - 2b_n = 3(b_n - 2b_{n-1}) + 2b_{n-1}.$$

Hence

$$b_{n+1} - 2b_n = 3b_n - 6b_{n-1} + 2b_{n-1} = 3b_n - 4b_{n-1},$$

so

$$b_{n+1} = 5b_n - 4b_{n-1}. \quad (3)$$

We now solve (3). Its characteristic equation is

$$r^2 - 5r + 4 = 0,$$

which factors as

$$(r - 4)(r - 1) = 0.$$

Thus

$$b_n = h_1 4^n + h_2.$$

We need two initial values. We already have

$$b_0 = 2.$$

Also,

$$b_1 = a_0 + 2b_0 = 1 + 2(2) = 5.$$

Therefore

$$2 = h_1 + h_2, \quad 5 = 4h_1 + h_2.$$

Subtracting gives

$$3 = 3h_1 \implies h_1 = 1,$$

and hence

$$h_2 = 1.$$

Therefore

$$b_n = 4^n + 1.$$

Finally, recover a_n from (2):

$$\begin{aligned} a_n &= b_{n+1} - 2b_n \\ &= (4^{n+1} + 1) - 2(4^n + 1) \\ &= 4 \cdot 4^n + 1 - 2 \cdot 4^n - 2 \\ &= 2 \cdot 4^n - 1. \end{aligned}$$

Thus

$$\boxed{a_n = 2 \cdot 4^n - 1, \quad b_n = 4^n + 1.}$$

Method 2: Direct eigenmode template

Let

$$\begin{pmatrix} a_n \\ b_n \end{pmatrix} = v r^n, \quad v \neq 0.$$

Then

$$v r^n = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix} v r^{n-1} \implies \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix} v = r v.$$

Hence

$$\det \begin{pmatrix} 3-r & 2 \\ 1 & 2-r \end{pmatrix} = (3-r)(2-r) - 2 = r^2 - 5r + 4 = (r-4)(r-1) = 0.$$

So the eigenmodes are

$$r = 4 \text{ with } v_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad r = 1 \text{ with } v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} a_n \\ b_n \end{pmatrix} = C_1 4^n \begin{pmatrix} 2 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Using

$$\begin{pmatrix} a_0 \\ b_0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} = C_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix},$$

we get

$$2C_1 - C_2 = 1, \quad C_1 + C_2 = 2,$$

so

$$C_1 = 1, \quad C_2 = 1.$$

Hence

$$\begin{pmatrix} a_n \\ b_n \end{pmatrix} = 4^n \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \cdot 4^n - 1 \\ 4^n + 1 \end{pmatrix}.$$

Thus

$$\boxed{a_n = 2 \cdot 4^n - 1, \quad b_n = 4^n + 1.}$$

Method 3: Matrix method

Write the system in vector form:

$$\begin{pmatrix} a_n \\ b_n \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a_{n-1} \\ b_{n-1} \end{pmatrix}.$$

Set

$$M = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}, \quad v_n = \begin{pmatrix} a_n \\ b_n \end{pmatrix}.$$

Then

$$v_n = Mv_{n-1} = M^n v_0, \quad v_0 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

We diagonalize M . Its characteristic polynomial is

$$\det(M - \lambda I) = \begin{vmatrix} 3 - \lambda & 2 \\ 1 & 2 - \lambda \end{vmatrix} = (3 - \lambda)(2 - \lambda) - 2 = \lambda^2 - 5\lambda + 4.$$

Hence the eigenvalues are

$$\lambda_1 = 4, \quad \lambda_2 = 1.$$

For $\lambda = 4$, solve

$$(M - 4I)x = 0 \implies \begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0.$$

So $x = 2y$, and one eigenvector is

$$v^{(1)} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

For $\lambda = 1$, solve

$$(M - I)x = 0 \implies \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0.$$

So $x = -y$, and one eigenvector is

$$v^{(2)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Express the initial vector as a linear combination:

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

This gives

$$\begin{cases} 2c_1 - c_2 = 1, \\ c_1 + c_2 = 2. \end{cases}$$

Solving yields

$$c_1 = 1, \quad c_2 = 1.$$

Therefore

$$v_n = M^n v_0 = 4^n \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \cdot 4^n - 1 \\ 4^n + 1 \end{pmatrix}.$$

Hence

$$\boxed{a_n = 2 \cdot 4^n - 1, \quad b_n = 4^n + 1.}$$

Question 2 (Ex. 8.2.11: Third-order homogeneous recurrence)

Problem

Find the solution to

$$a_n = 2a_{n-1} + 5a_{n-2} - 6a_{n-3},$$

with

$$a_0 = 7, \quad a_1 = -4, \quad a_2 = 8.$$

Solution

Method 1: Characteristic equation

The recurrence is linear homogeneous of degree 3. Its characteristic equation is

$$r^3 - 2r^2 - 5r + 6 = 0.$$

We factor the polynomial:

$$r^3 - 2r^2 - 5r + 6 = (r - 1)(r + 2)(r - 3).$$

So the three distinct roots are

$$r = 1, \quad r = -2, \quad r = 3.$$

Hence the general solution has the form

$$a_n = h_1 \cdot 1^n + h_2(-2)^n + h_3 3^n = h_1 + h_2(-2)^n + h_3 3^n.$$

Now use the initial conditions:

$$a_0 = 7 \implies h_1 + h_2 + h_3 = 7, \tag{1}$$

$$a_1 = -4 \implies h_1 - 2h_2 + 3h_3 = -4, \tag{2}$$

$$a_2 = 8 \implies h_1 + 4h_2 + 9h_3 = 8. \tag{3}$$

Subtract (1) from (2):

$$-3h_2 + 2h_3 = -11. \tag{4}$$

Subtract (1) from (3):

$$3h_2 + 8h_3 = 1. \tag{5}$$

Add (4) and (5):

$$10h_3 = -10 \implies h_3 = -1.$$

Then from (5),

$$3h_2 + 8(-1) = 1 \implies 3h_2 = 9 \implies h_2 = 3.$$

Finally, from (1),

$$h_1 + 3 - 1 = 7 \implies h_1 = 5.$$

Therefore

$$\boxed{a_n = 5 + 3(-2)^n - 3^n.}$$

Method 2: Direct characteristic-root template

$$r^3 - 2r^2 - 5r + 6 = (r - 1)(r + 2)(r - 3) = 0.$$

Distinct roots $1, -2, 3$. Hence

$$a_n = C_1 + C_2(-2)^n + C_33^n.$$

Using $a_0 = 7$, $a_1 = -4$, $a_2 = 8$,

$$C_1 + C_2 + C_3 = 7,$$

$$C_1 - 2C_2 + 3C_3 = -4,$$

$$C_1 + 4C_2 + 9C_3 = 8.$$

Hence

$$-3C_2 + 2C_3 = -11, \quad 3C_2 + 8C_3 = 1.$$

So

$$C_3 = -1, \quad C_2 = 3, \quad C_1 = 5.$$

Therefore

$$\boxed{a_n = 5 + 3(-2)^n - 3^n.}$$

Method 3: Generating function

Let

$$A(x) = \sum_{n \geq 0} a_n x^n.$$

For $n \geq 3$, the recurrence gives

$$a_n - 2a_{n-1} - 5a_{n-2} + 6a_{n-3} = 0.$$

Multiply by x^n and sum for $n \geq 3$:

$$\sum_{n \geq 3} a_n x^n - 2 \sum_{n \geq 3} a_{n-1} x^n - 5 \sum_{n \geq 3} a_{n-2} x^n + 6 \sum_{n \geq 3} a_{n-3} x^n = 0.$$

Rewrite each sum:

$$\sum_{n \geq 3} a_n x^n = A(x) - a_0 - a_1 x - a_2 x^2 = A(x) - 7 + 4x - 8x^2,$$

$$\sum_{n \geq 3} a_{n-1} x^n = x \sum_{n \geq 3} a_{n-1} x^{n-1} = x(A(x) - a_0 - a_1 x) = x(A(x) - 7 + 4x),$$

$$\sum_{n \geq 3} a_{n-2} x^n = x^2(A(x) - a_0) = x^2(A(x) - 7),$$

$$\sum_{n \geq 3} a_{n-3} x^n = x^3 A(x).$$

Substitute these into the recurrence sum:

$$(A(x) - 7 + 4x - 8x^2) - 2x(A(x) - 7 + 4x) - 5x^2(A(x) - 7) + 6x^3 A(x) = 0.$$

Collect terms:

$$A(x)(1 - 2x - 5x^2 + 6x^3) - 7 + 18x + 13x^2 = 0.$$

Hence

$$A(x) = \frac{7 - 18x - 13x^2}{1 - 2x - 5x^2 + 6x^3}.$$

Factor the denominator:

$$1 - 2x - 5x^2 + 6x^3 = (1 - x)(1 + 2x)(1 - 3x).$$

So

$$A(x) = \frac{5}{1 - x} + \frac{3}{1 + 2x} - \frac{1}{1 - 3x}.$$

Therefore

$$A(x) = 5 \sum_{n \geq 0} x^n + 3 \sum_{n \geq 0} (-2x)^n - \sum_{n \geq 0} (3x)^n = \sum_{n \geq 0} (5 + 3(-2)^n - 3^n) x^n.$$

Thus

$$\boxed{a_n = 5 + 3(-2)^n - 3^n.}$$

Question 3 (Ex. 8.2.15: Repeated root of multiplicity three)

Problem

Solve the recurrence relation

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3},$$

with

$$a_0 = 5, \quad a_1 = -9, \quad a_2 = 15.$$

Solution

Method 1: Characteristic equation

The characteristic equation is

$$r^3 + 3r^2 + 3r + 1 = 0.$$

This factors as

$$(r + 1)^3 = 0.$$

Hence $r = -1$ is a root of multiplicity 3. Therefore the general solution is

$$a_n = (h_1 n^2 + h_2 n + h_3)(-1)^n.$$

Use the initial conditions:

$$a_0 = 5 \implies h_3 = 5.$$

Next,

$$a_1 = -9 \implies -(h_1 + h_2 + h_3) = -9 \implies h_1 + h_2 + 5 = 9 \implies h_1 + h_2 = 4. \quad (1)$$

Also,

$$a_2 = 15 \implies (4h_1 + 2h_2 + 5) = 15 \implies 2h_1 + h_2 = 5. \quad (2)$$

Subtract (1) from (2):

$$h_1 = 1.$$

Then from (1),

$$h_2 = 3.$$

Thus

$$\boxed{a_n = (n^2 + 3n + 5)(-1)^n.}$$

Method 2: Direct repeated-root template

$$r^3 + 3r^2 + 3r + 1 = (r + 1)^3 = 0.$$

Root -1 has multiplicity 3. Hence

$$a_n = (C_1 + C_2n + C_3n^2)(-1)^n.$$

Using $a_0 = 5$,

$$C_1 = 5.$$

Using $a_1 = -9$,

$$-(C_1 + C_2 + C_3) = -9 \implies C_2 + C_3 = 4.$$

Using $a_2 = 15$,

$$C_1 + 2C_2 + 4C_3 = 15 \implies C_2 + 2C_3 = 5.$$

Hence

$$C_3 = 1, \quad C_2 = 3.$$

Therefore

$$\boxed{a_n = (n^2 + 3n + 5)(-1)^n.}$$

Method 3: Remove the alternating sign

Define

$$b_n = (-1)^n a_n.$$

Then $a_n = (-1)^n b_n$. Substitute this into the recurrence:

$$(-1)^n b_n = -3(-1)^{n-1} b_{n-1} - 3(-1)^{n-2} b_{n-2} - (-1)^{n-3} b_{n-3}.$$

Factor out $(-1)^n$:

$$(-1)^n b_n = (-1)^n (3b_{n-1} - 3b_{n-2} + b_{n-3}).$$

So

$$b_n = 3b_{n-1} - 3b_{n-2} + b_{n-3}.$$

Equivalently,

$$b_n - 3b_{n-1} + 3b_{n-2} - b_{n-3} = 0.$$

But this is precisely the condition that the third forward difference of b_n is zero. Therefore b_n must be a polynomial of degree at most 2:

$$b_n = An^2 + Bn + C.$$

Now compute the transformed initial values:

$$b_0 = (-1)^0 a_0 = 5, \quad b_1 = (-1)^1 a_1 = 9, \quad b_2 = (-1)^2 a_2 = 15.$$

Hence

$$C = 5,$$

$$A + B + 5 = 9 \implies A + B = 4, \tag{1}$$

$$4A + 2B + 5 = 15 \implies 2A + B = 5. \tag{2}$$

Subtracting (1) from (2) gives

$$A = 1,$$

and then

$$B = 3.$$

So

$$b_n = n^2 + 3n + 5.$$

Therefore

$$a_n = (-1)^n b_n = (n^2 + 3n + 5)(-1)^n.$$

Thus

$$\boxed{a_n = (n^2 + 3n + 5)(-1)^n.}$$

Question 4 (Ex. 8.2.22: A first-order non-homogeneous recurrence)

Problem

- (a) Find all solutions of

$$a_n = 2a_{n-1} + 2n^2 \quad (n \geq 2).$$

- (b) Find the solution of the recurrence relation in part (a) with initial condition $a_1 = 4$.

Solution

Method 1: Undetermined coefficients

- (a) First solve the associated homogeneous recurrence

$$b_n = 2b_{n-1}.$$

Its general solution is

$$b_n = h 2^n.$$

We now seek a particular solution of the original recurrence

$$a_n = 2a_{n-1} + 2n^2.$$

Since the forcing term $2n^2$ is a polynomial of degree 2, and 1 is not a root of the homogeneous characteristic equation $r - 2 = 0$, we try a particular solution of the form

$$a_n^{(p)} = cn^2 + dn + e.$$

Substitute:

$$cn^2 + dn + e = 2(c(n-1)^2 + d(n-1) + e) + 2n^2.$$

Expand the right-hand side:

$$\begin{aligned} 2(c(n-1)^2 + d(n-1) + e) + 2n^2 &= 2(cn^2 - 2cn + c + dn - d + e) + 2n^2 \\ &= (2c + 2)n^2 + (-4c + 2d)n + (2c - 2d + 2e). \end{aligned}$$

Hence coefficient comparison gives

$$c = 2c + 2, \quad d = -4c + 2d, \quad e = 2c - 2d + 2e.$$

So

$$c = -2, \quad d = -8, \quad e = -12.$$

Thus a particular solution is

$$a_n^{(p)} = -2n^2 - 8n - 12.$$

Therefore the general solution is

$$a_n = h 2^n - 2n^2 - 8n - 12.$$

Hence

$$a_n = h 2^n - 2n^2 - 8n - 12 \quad (n \geq 2), \text{ for arbitrary constant } h.$$

(b) Use $a_1 = 4$:

$$4 = h \cdot 2^1 - 2(1)^2 - 8(1) - 12 = 2h - 22.$$

Hence

$$2h = 26 \implies h = 13.$$

Therefore

$$a_n = 13 \cdot 2^n - 2n^2 - 8n - 12 \quad (n \geq 1).$$

Method 2: Direct template from the non-homogeneous form

(a) Homogeneous part:

$$a_n^{(h)} = C 2^n.$$

Since the forcing term is a degree-2 polynomial and 1 is not a root of $r - 2 = 0$, take

$$a_n^{(p)} = An^2 + Bn + D.$$

Substitute into

$$\begin{aligned} a_n &= 2a_{n-1} + 2n^2 : \\ An^2 + Bn + D &= 2(A(n-1)^2 + B(n-1) + D) + 2n^2. \end{aligned}$$

Hence

$$An^2 + Bn + D = (2A + 2)n^2 + (-4A + 2B)n + (2A - 2B + 2D).$$

So

$$A = 2A + 2, \quad B = -4A + 2B, \quad D = 2A - 2B + 2D.$$

Thus

$$A = -2, \quad B = -8, \quad D = -12.$$

Therefore

$$a_n = C 2^n - 2n^2 - 8n - 12.$$

Hence

$$\boxed{a_n = C 2^n - 2n^2 - 8n - 12 \quad (n \geq 2).}$$

(b) Using $a_1 = 4$,

$$4 = 2C - 2 - 8 - 12 = 2C - 22.$$

So

$$C = 13.$$

Hence

$$\boxed{a_n = 13 \cdot 2^n - 2n^2 - 8n - 12 \quad (n \geq 1).}$$

Method 3: Transform the sequence to remove the forcing term

(a) We look for a quadratic polynomial $p(n)$ such that

$$c_n := a_n + p(n)$$

satisfies a homogeneous recurrence of the form

$$c_n = 2c_{n-1}.$$

This requires

$$a_n + p(n) = 2(a_{n-1} + p(n-1)).$$

Using the given recurrence $a_n = 2a_{n-1} + 2n^2$, this becomes

$$2a_{n-1} + 2n^2 + p(n) = 2a_{n-1} + 2p(n-1).$$

Hence we need

$$p(n) - 2p(n-1) = -2n^2. \quad (1)$$

Take

$$p(n) = 2n^2 + 8n + 12.$$

Then

$$\begin{aligned} p(n) - 2p(n-1) &= (2n^2 + 8n + 12) - 2(2(n-1)^2 + 8(n-1) + 12) \\ &= 2n^2 + 8n + 12 - 2(2n^2 - 4n + 2 + 8n - 8 + 12) \\ &= 2n^2 + 8n + 12 - (4n^2 + 8n + 12) \\ &= -2n^2. \end{aligned}$$

So (1) holds.

Therefore, if we define

$$c_n = a_n + 2n^2 + 8n + 12,$$

then

$$c_n = 2c_{n-1}.$$

Hence

$$c_n = h 2^n$$

for some constant h . Returning to a_n ,

$$a_n = h 2^n - 2n^2 - 8n - 12.$$

So

$$\boxed{a_n = h 2^n - 2n^2 - 8n - 12 \quad (n \geq 2).}$$

(b) Using $a_1 = 4$,

$$c_1 = a_1 + 2(1)^2 + 8(1) + 12 = 4 + 2 + 8 + 12 = 26.$$

But also

$$c_1 = h \cdot 2^1 = 2h.$$

So

$$h = 13.$$

Hence

$$\boxed{a_n = 13 \cdot 2^n - 2n^2 - 8n - 12 \quad (n \geq 1).}$$

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Question 5 (Ex. 8.2.25 modified: Recurrences with different forcing terms)

Problem

Find all solutions of the recurrence relation

$$a_n = 4a_{n-1} - 3a_{n-2} + F(n),$$

where $F(n)$ is

- (a) 2^n ,
- (b) 3^n ,
- (c) $n + 3$,
- (d) $n + 2^n + 3$. *Hint: To obtain a particular solution for $n + 2^n + 3$, add up the particular solutions in Parts (a) and (c).*

Solution

Method 1: Solve the homogeneous part, then use undetermined coefficients

The associated homogeneous recurrence is

$$b_n = 4b_{n-1} - 3b_{n-2}.$$

Its characteristic equation is

$$r^2 - 4r + 3 = 0 = (r - 1)(r - 3).$$

Thus the homogeneous solution is

$$b_n = h_1 + h_2 3^n.$$

- (a) Here $F(n) = 2^n$. Since 2 is not a root of the homogeneous characteristic equation, try

$$a_n^{(p)} = c 2^n.$$

Substitute:

$$c 2^n = 4c 2^{n-1} - 3c 2^{n-2} + 2^n.$$

Divide by 2^{n-2} :

$$4c = 8c - 3c + 4.$$

Hence

$$-c = 4 \implies c = -4.$$

So

$$a_n^{(p)} = -4 \cdot 2^n = -2^{n+2}.$$

Therefore

$$\boxed{a_n = h_1 + h_2 3^n - 2^{n+2}.$$

- (b) Here $F(n) = 3^n$. Since 3 is a root of the homogeneous characteristic equation and it has multiplicity 1, try

$$a_n^{(p)} = c n 3^n.$$

Substitute:

$$cn3^n = 4c(n-1)3^{n-1} - 3c(n-2)3^{n-2} + 3^n.$$

Divide by 3^{n-2} :

$$9cn = 12c(n-1) - 3c(n-2) + 9.$$

Simplify the right-hand side:

$$12c(n-1) - 3c(n-2) = 12cn - 12c - 3cn + 6c = 9cn - 6c.$$

So

$$9cn = 9cn - 6c + 9,$$

hence

$$6c = 9 \implies c = \frac{3}{2}.$$

Therefore

$$a_n^{(p)} = \frac{3}{2} n 3^n = \frac{n 3^{n+1}}{2}.$$

Thus

$$a_n = h_1 + h_2 3^n + \frac{n 3^{n+1}}{2}.$$

- (c) Here $F(n) = n+3$, a polynomial of degree 1. Since 1 is a root of the homogeneous characteristic equation and has multiplicity 1, we multiply the usual degree-1 polynomial trial by n . So try

$$a_n^{(p)} = n(cn + d) = cn^2 + dn.$$

Substitute:

$$cn^2 + dn = 4(c(n-1)^2 + d(n-1)) - 3(c(n-2)^2 + d(n-2)) + n + 3.$$

Expand:

$$\begin{aligned} 4(c(n-1)^2 + d(n-1)) &= 4(cn^2 - 2cn + c + dn - d) \\ &= 4cn^2 + (-8c + 4d)n + (4c - 4d), \end{aligned}$$

and

$$\begin{aligned} -3(c(n-2)^2 + d(n-2)) &= -3(cn^2 - 4cn + 4c + dn - 2d) \\ &= -3cn^2 + (12c - 3d)n + (-12c + 6d). \end{aligned}$$

Adding $n + 3$, the right-hand side becomes

$$cn^2 + (4c + d + 1)n + (-8c + 2d + 3).$$

Therefore

$$cn^2 + dn = cn^2 + (4c + d + 1)n + (-8c + 2d + 3).$$

Comparing coefficients yields

$$d = 4c + d + 1, \quad 0 = -8c + 2d + 3.$$

Thus

$$4c + 1 = 0 \implies c = -\frac{1}{4},$$

and then

$$0 = -8 \left(-\frac{1}{4} \right) + 2d + 3 = 2 + 2d + 3 \implies 2d = -5 \implies d = -\frac{5}{2}.$$

So

$$a_n^{(p)} = -\frac{n^2}{4} - \frac{5n}{2}.$$

Therefore

$$a_n = h_1 + h_2 3^n - \frac{n^2}{4} - \frac{5n}{2}.$$

- (d) Here $F(n) = n + 2^n + 3$. By linearity, we may add the particular solutions from parts (a) and (c):

$$a_n^{(p)} = -2^{n+2} - \frac{n^2}{4} - \frac{5n}{2}.$$

Hence

$$a_n = h_1 + h_2 3^n - 2^{n+2} - \frac{n^2}{4} - \frac{5n}{2}.$$

Method 2: Direct trial-template method

$$r^2 - 4r + 3 = (r - 1)(r - 3) = 0 \implies a_n^{(h)} = C_1 + C_2 3^n.$$

(a) $F(n) = 2^n$. Since $2 \notin \{1, 3\}$, take

$$a_n^{(p)} = A 2^n.$$

Then

$$A 2^n = 4A 2^{n-1} - 3A 2^{n-2} + 2^n \implies 4A = 8A - 3A + 4,$$

so

$$A = -4.$$

Hence

$$\boxed{a_n = C_1 + C_2 3^n - 2^{n+2}}.$$

(b) $F(n) = 3^n$. Since 3 is a simple homogeneous root, take

$$a_n^{(p)} = A n 3^n.$$

Then

$$9A n = 12A(n - 1) - 3A(n - 2) + 9 = 9A n - 6A + 9,$$

so

$$A = \frac{3}{2}.$$

Hence

$$\boxed{a_n = C_1 + C_2 3^n + \frac{n 3^{n+1}}{2}}.$$

(c) $F(n) = n + 3$. Polynomial of degree 1, and 1 is a simple homogeneous root, so take

$$a_n^{(p)} = n(An + B) = An^2 + Bn.$$

Substituting,

$$An^2 + Bn = An^2 + (4A + B + 1)n + (-8A + 2B + 3).$$

Hence

$$4A + 1 = 0, \quad -8A + 2B + 3 = 0,$$

so

$$A = -\frac{1}{4}, \quad B = -\frac{5}{2}.$$

Therefore

$$\boxed{a_n = C_1 + C_2 3^n - \frac{n^2}{4} - \frac{5n}{2}}.$$

(d) $F(n) = n + 2^n + 3$. Add the particular solutions from (a) and (c):

$$a_n^{(p)} = -2^{n+2} - \frac{n^2}{4} - \frac{5n}{2}.$$

Thus

$$\boxed{a_n = C_1 + C_2 3^n - 2^{n+2} - \frac{n^2}{4} - \frac{5n}{2}}.$$

Method 3: Annihilator/superposition viewpoint

Again the homogeneous solution is

$$b_n = h_1 + h_2 3^n.$$

- (a) Since the forcing term is 2^n , an annihilator is $E - 2$, where E is the shift operator $(Ea)_n = a_{n+1}$. The homogeneous operator is

$$E^2 - 4E + 3 = (E - 1)(E - 3).$$

Because 2 is not a root of the homogeneous characteristic polynomial, the full solution space must contain a term proportional to 2^n in addition to the homogeneous part:

$$a_n = h_1 + h_2 3^n + c 2^n.$$

Substituting this form into the recurrence determines $c = -4$, as in Method 1. Therefore

$$a_n = h_1 + h_2 3^n - 2^{n+2}.$$

- (b) Since the forcing term is 3^n , whose base 3 is already a simple homogeneous root, the annihilator method predicts a particular term of the form

$$cn 3^n.$$

Thus

$$a_n = h_1 + h_2 3^n + cn 3^n.$$

Substituting into the recurrence yields $c = \frac{3}{2}$, so

$$a_n = h_1 + h_2 3^n + \frac{n 3^{n+1}}{2}.$$

- (c) Since the forcing term $n + 3$ is a polynomial of degree 1, its annihilator is $(E - 1)^2$. Because 1 is already a simple homogeneous root, the particular solution must be of the form

$$cn^2 + dn.$$

So

$$a_n = h_1 + h_2 3^n + cn^2 + dn.$$

Substitution gives

$$c = -\frac{1}{4}, \quad d = -\frac{5}{2}.$$

Hence

$$a_n = h_1 + h_2 3^n - \frac{n^2}{4} - \frac{5n}{2}.$$

(d) By linearity of the recurrence operator

$$L(a)_n := a_n - 4a_{n-1} + 3a_{n-2},$$

if p_n is a particular solution for 2^n and q_n is a particular solution for $n+3$, then $p_n + q_n$ is a particular solution for $n + 2^n + 3$. Using parts (a) and (c),

$$p_n = -2^{n+2}, \quad q_n = -\frac{n^2}{4} - \frac{5n}{2},$$

so

$$p_n + q_n = -2^{n+2} - \frac{n^2}{4} - \frac{5n}{2}.$$

Therefore

$$a_n = h_1 + h_2 3^n - 2^{n+2} - \frac{n^2}{4} - \frac{5n}{2}.$$

Question 6 (Ex. 8.2.26: Sum of the first n perfect squares)

Problem

Let

$$a_n = \sum_{k=1}^n k^2.$$

Using linear non-homogeneous recurrence relations, determine a formula for a_n .

Solution

Method 1: Solve the recurrence by undetermined coefficients

Since

$$a_n = a_{n-1} + n^2 \quad (n \geq 1),$$

with

$$a_0 = 0,$$

we solve the non-homogeneous recurrence

$$a_n - a_{n-1} = n^2.$$

The associated homogeneous recurrence is

$$b_n = b_{n-1},$$

whose general solution is a constant:

$$b_n = \alpha.$$

Because the forcing term is a polynomial of degree 2, and 1 is a root of the homogeneous characteristic equation of multiplicity 1, we seek a particular solution of the form

$$a_n^{(p)} = cn^3 + dn^2 + en.$$

Substitute into

$$a_n = a_{n-1} + n^2 :$$

$$cn^3 + dn^2 + en = c(n-1)^3 + d(n-1)^2 + e(n-1) + n^2.$$

Expand the right-hand side:

$$\begin{aligned} c(n-1)^3 + d(n-1)^2 + e(n-1) + n^2 &= c(n^3 - 3n^2 + 3n - 1) + d(n^2 - 2n + 1) + en - e + n^2 \\ &= cn^3 + (-3c + d + 1)n^2 + (3c - 2d + e)n + (-c + d - e). \end{aligned}$$

Matching coefficients with the left-hand side $cn^3 + dn^2 + en$ gives

$$d = -3c + d + 1, \quad e = 3c - 2d + e, \quad 0 = -c + d - e.$$

Hence

$$3c = 1 \implies c = \frac{1}{3},$$

$$0 = 3c - 2d \implies d = \frac{1}{2},$$

$$0 = -c + d - e \implies e = \frac{1}{6}.$$

Therefore

$$a_n^{(p)} = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.$$

The general solution is

$$a_n = \alpha + \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.$$

Use $a_0 = 0$:

$$0 = \alpha.$$

Hence

$$a_n = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.$$

Put everything over the common denominator 6:

$$a_n = \frac{2n^3 + 3n^2 + n}{6} = \frac{n(2n^2 + 3n + 1)}{6} = \frac{n(n+1)(2n+1)}{6}.$$

Therefore

$$a_n = \frac{n(n+1)(2n+1)}{6}.$$

Method 2: Direct polynomial-template method

$$a_n - a_{n-1} = n^2, \quad a_0 = 0.$$

Homogeneous part:

$$a_n^{(h)} = C.$$

Since the forcing term is degree 2 and 1 is a simple homogeneous root, take

$$a_n^{(p)} = An^3 + Bn^2 + Cn.$$

Then

$$An^3 + Bn^2 + Cn = A(n-1)^3 + B(n-1)^2 + C(n-1) + n^2.$$

So

$$An^3 + Bn^2 + Cn = An^3 + (-3A + B + 1)n^2 + (3A - 2B + C)n + (-A + B - C).$$

Hence

$$3A = 1, \quad 3A - 2B = 0, \quad -A + B - C = 0.$$

Thus

$$A = \frac{1}{3}, \quad B = \frac{1}{2}, \quad C = \frac{1}{6}.$$

Therefore

$$a_n = K + \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.$$

Using $a_0 = 0$,

$$K = 0.$$

Hence

$$a_n = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n = \frac{n(n+1)(2n+1)}{6}.$$

Thus

$$\boxed{a_n = \frac{n(n+1)(2n+1)}{6}}.$$

Method 3: Generating function for the recurrence

Again we use

$$a_n = a_{n-1} + n^2, \quad a_0 = 0.$$

Let

$$A(x) = \sum_{n \geq 0} a_n x^n.$$

Multiply the recurrence by x^n and sum for $n \geq 1$:

$$\sum_{n \geq 1} a_n x^n = \sum_{n \geq 1} a_{n-1} x^n + \sum_{n \geq 1} n^2 x^n.$$

Now

$$\sum_{n \geq 1} a_n x^n = A(x),$$

because $a_0 = 0$, and

$$\sum_{n \geq 1} a_{n-1} x^n = xA(x).$$

Also, the standard generating function for squares is

$$\sum_{n \geq 1} n^2 x^n = \frac{x(x+1)}{(1-x)^3}.$$

Hence

$$A(x) = xA(x) + \frac{x(x+1)}{(1-x)^3}.$$

So

$$(1-x)A(x) = \frac{x(x+1)}{(1-x)^3},$$

and therefore

$$A(x) = \frac{x(x+1)}{(1-x)^4}.$$

Now use

$$(1-x)^{-4} = \sum_{n \geq 0} \binom{n+3}{3} x^n.$$

Thus

$$A(x) = x(1+x)(1-x)^{-4} = x(1-x)^{-4} + x^2(1-x)^{-4}.$$

So the coefficient of x^n is

$$a_n = \binom{n+2}{3} + \binom{n+1}{3}.$$

Compute:

$$\begin{aligned}a_n &= \frac{(n+2)(n+1)n}{6} + \frac{(n+1)n(n-1)}{6} \\&= \frac{n(n+1)((n+2) + (n-1))}{6} \\&= \frac{n(n+1)(2n+1)}{6}.\end{aligned}$$

Hence

$$a_n = \frac{n(n+1)(2n+1)}{6}.$$

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Question 7 (Additional Exercise 2: A higher-order recurrence for Fibonacci numbers)

Problem

Show that the Fibonacci numbers satisfy the recurrence relation

$$f_n = 5f_{n-4} + 3f_{n-5} \quad (n = 5, 6, 7, \dots),$$

together with the initial conditions

$$f_0 = 0, \quad f_1 = 1, \quad f_2 = 1, \quad f_3 = 2, \quad f_4 = 3.$$

Solution

Method 1: Repeated substitution from the Fibonacci recurrence

The Fibonacci recurrence is

$$f_n = f_{n-1} + f_{n-2}. \quad (1)$$

Also,

$$f_{n-1} = f_{n-2} + f_{n-3}. \quad (2)$$

Substitute (2) into (1):

$$f_n = (f_{n-2} + f_{n-3}) + f_{n-2} = 2f_{n-2} + f_{n-3}. \quad (3)$$

Next,

$$f_{n-2} = f_{n-3} + f_{n-4}. \quad (4)$$

Substitute (4) into (3):

$$f_n = 2(f_{n-3} + f_{n-4}) + f_{n-3} = 3f_{n-3} + 2f_{n-4}. \quad (5)$$

Now use

$$f_{n-3} = f_{n-4} + f_{n-5}. \quad (6)$$

Substitute (6) into (5):

$$f_n = 3(f_{n-4} + f_{n-5}) + 2f_{n-4} = 5f_{n-4} + 3f_{n-5}.$$

Thus

$$\boxed{f_n = 5f_{n-4} + 3f_{n-5} \quad (n \geq 5).}$$

The initial conditions are the usual Fibonacci initial values:

$$\boxed{f_0 = 0, \quad f_1 = 1, \quad f_2 = 1, \quad f_3 = 2, \quad f_4 = 3.}$$

Method 2: Direct closed-form/root check

The Fibonacci characteristic equation is

$$r^2 - r - 1 = 0.$$

Let its roots be

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

Then

$$f_n = \frac{\phi^n - \psi^n}{\sqrt{5}}.$$

Since $r^2 = r + 1$, we get

$$r^3 = 2r + 1, \quad r^4 = 3r + 2, \quad r^5 = 5r + 3.$$

Hence each of $r = \phi, \psi$ satisfies

$$r^5 - 5r - 3 = 0.$$

Therefore

$$\phi^n = 5\phi^{n-4} + 3\phi^{n-5}, \quad \psi^n = 5\psi^{n-4} + 3\psi^{n-5}.$$

Subtracting and dividing by $\sqrt{5}$,

$$f_n = 5f_{n-4} + 3f_{n-5} \quad (n \geq 5).$$

Also

$$f_0 = 0, \quad f_1 = 1, \quad f_2 = 1, \quad f_3 = 2, \quad f_4 = 3.$$

Thus

$$\boxed{f_n = 5f_{n-4} + 3f_{n-5} \quad (n \geq 5).}$$

Method 3: Operator factorization

Let E denote the shift operator:

$$(Ef)_n = f_{n+1}.$$

The Fibonacci recurrence

$$f_n = f_{n-1} + f_{n-2}$$

can be written as

$$(E^2 - E - 1)f = 0.$$

Now factor the polynomial

$$E^5 - 5E - 3.$$

Direct multiplication gives

$$(E^2 - E - 1)(E^3 + E^2 + 2E + 3) = E^5 - 5E - 3.$$

Since $(E^2 - E - 1)f = 0$, applying $E^3 + E^2 + 2E + 3$ to both sides still yields zero, so

$$(E^5 - 5E - 3)f = 0.$$

This means

$$f_{n+5} - 5f_{n+1} - 3f_n = 0.$$

Reindex by replacing n with $n - 5$. Then for $n \geq 5$,

$$f_n - 5f_{n-4} - 3f_{n-5} = 0,$$

that is,

$$\boxed{f_n = 5f_{n-4} + 3f_{n-5} \quad (n \geq 5).}$$

The initial conditions remain

$$\boxed{f_0 = 0, \quad f_1 = 1, \quad f_2 = 1, \quad f_3 = 2, \quad f_4 = 3.}$$

Question 8 (Additional Exercise 3: Linear combinations of solutions)

Problem

Suppose that the sequences $(s_0, s_1, s_2, \dots)$ and $(t_0, t_1, t_2, \dots)$ both satisfy the same linear homogeneous recurrence relation with constant coefficients:

$$s_k = 5s_{k-1} - 4s_{k-2}, \quad t_k = 5t_{k-1} - 4t_{k-2} \quad (k \geq 2).$$

Define a new sequence $(u_0, u_1, u_2, \dots)$ by

$$u_i = 2s_i + 3t_i.$$

Show that $(u_0, u_1, u_2, \dots)$ also satisfies the same relation, that is,

$$u_k = 5u_{k-1} - 4u_{k-2}.$$

Solution

Method 1: Direct substitution

From the hypotheses,

$$s_k = 5s_{k-1} - 4s_{k-2}, \quad t_k = 5t_{k-1} - 4t_{k-2}.$$

Multiply the first equation by 2:

$$2s_k = 10s_{k-1} - 8s_{k-2}. \tag{1}$$

Multiply the second equation by 3:

$$3t_k = 15t_{k-1} - 12t_{k-2}. \tag{2}$$

Add (1) and (2):

$$2s_k + 3t_k = 10s_{k-1} + 15t_{k-1} - 8s_{k-2} - 12t_{k-2}.$$

Group the terms:

$$2s_k + 3t_k = 5(2s_{k-1} + 3t_{k-1}) - 4(2s_{k-2} + 3t_{k-2}).$$

By the definition $u_i = 2s_i + 3t_i$, this becomes

$$u_k = 5u_{k-1} - 4u_{k-2}.$$

Therefore

$$\boxed{u_k = 5u_{k-1} - 4u_{k-2} \quad (k \geq 2).}$$

Method 2: Direct closed-form template

The common recurrence has characteristic equation

$$r^2 - 5r + 4 = (r - 1)(r - 4) = 0.$$

Hence

$$s_k = A + B4^k, \quad t_k = C + D4^k$$

for some constants A, B, C, D . Therefore

$$u_k = 2s_k + 3t_k = 2(A + B4^k) + 3(C + D4^k) = (2A + 3C) + (2B + 3D)4^k.$$

So u_k is again of the form

$$u_k = E + F4^k,$$

which is exactly the general solution of

$$u_k = 5u_{k-1} - 4u_{k-2}.$$

Hence

$$\boxed{u_k = 5u_{k-1} - 4u_{k-2} \quad (k \geq 2).}$$

Method 3: Use linearity of the recurrence operator

Define the linear operator

$$L(x)_k := x_k - 5x_{k-1} + 4x_{k-2} \quad (k \geq 2).$$

The assumptions that (s_k) and (t_k) satisfy the recurrence are exactly

$$L(s)_k = 0, \quad L(t)_k = 0 \quad (k \geq 2).$$

Now let

$$u_k = 2s_k + 3t_k.$$

Because L is linear,

$$L(u)_k = L(2s + 3t)_k = 2L(s)_k + 3L(t)_k.$$

But $L(s)_k = 0$ and $L(t)_k = 0$, so

$$L(u)_k = 2 \cdot 0 + 3 \cdot 0 = 0.$$

Hence

$$u_k - 5u_{k-1} + 4u_{k-2} = 0,$$

or equivalently,

$$\boxed{u_k = 5u_{k-1} - 4u_{k-2} \quad (k \geq 2).}$$